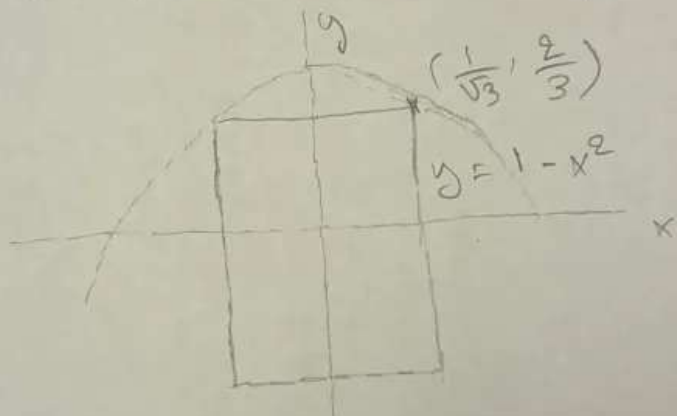


Problem 1: so R got corners and it also has width $2a$ height $2b$ and $b = 1 - a^2$ and $0 \leq a \leq 1$ so $A(a) = 4ab = 4a(1 - a^2)$ so if we have $A'(a) = 4 - 12a^2 = 0$, giving $a = \frac{1}{\sqrt{3}}$ and $A'' < 0$ so it's a max so then $b = \frac{2}{3}$ and $A_{\max} = \frac{8\sqrt{3}}{9}$ so the sides are $2a = \frac{2\sqrt{3}}{3} \approx 1.155$ and $2b = \frac{4}{3} \approx 1.333$ so it's not a square, it's a bit taller and wider



2) If it's falling from rest, $v^2 = 2gh$: $v = \sqrt{2(9.8)(200)}$
 $= \sqrt{3920} = 28\sqrt{10} \approx 88.54 \text{ m/s}$ and that's under 90 so yes it survives!

3) Differentiating $\int_{-x}^x f(t) dt = 0$ gives $f(x) + f(-x) = 0$ so f has to be odd. Two examples $f(t) = t$ and $f(t) = \sin t$

ii) Need to do some calculation as (i) so condition is $f(x) + f(-x) = 0$ so basically it's any continuous odd func

iii) there's not a function like that other than $f \equiv 0$

but if $G(x) = \int_{-x}^{2x} f$ then if $G \equiv 0$ then

$$G'(x) = 2f(2x) + f(-x) = 0 \text{ and then we plug in}$$

$$x = 0 \text{ so } 3f(0) = 0 \text{ so } f(0) = 0 \text{ and swapping}$$

$$x \rightarrow -x \text{ gives } f(x) = -2f(-2x) \text{ and using the same}$$

$$\text{relation on } f(-2x) \text{ gives } f(-2x) = -2f(4x) \text{ so}$$

$$\text{combining } f(x) = 4f(4x), f\left(\frac{x}{4^n}\right) = 4^n f(x)$$

As $n \rightarrow \infty$ the left side $\rightarrow f(0) = 0$ so by continuity $4^n f(x) \rightarrow 0$ which forces $f(x) = 0$ so there's non constant continuous example exists

iv) a) Differentiating both sides of $\frac{\pi}{2} + \int_a^x f = x \sin x$ gives $f(x) = \frac{d}{dx}(x \sin x) = \sin x + x \cos x$

and if we plug $x = a$ into the original equation won't work so instead $a = \frac{\pi}{2}$ so $\frac{\pi}{2} \sin \frac{\pi}{2}$

$$\text{so } \boxed{f(x) = \sin x + x \cos x, a = \frac{\pi}{2}} \quad = \frac{\pi}{2} \cdot 1$$

5) a) So I differentiate $xy = 1 : y + xy' = 0$, so
 $y' = -y/x = -1/a^2$ at $(a, 1/a)$ So the perpendicular
 slope is the reciprocal $m_{\perp} = a^2$
 negative

b) ~~Let~~ $P = (x, 1/x)$ and minimize the squared
 distance I get $Q = (0, 2) \Rightarrow D^2(x) = x^2 + (1/x - 2)^2$
 So at closest point, $\frac{d}{dx} D^2 = 0$ So:

$$2x + 2\left(\frac{1}{x} - 2\right)\left(-\frac{1}{x^2}\right) = 0 \Rightarrow x^4 + 2x - 1 = 0$$

So now the slope of QP is $m_{QP} =$

$$\frac{1/x - 2}{x} = \frac{1 - 2x}{x^2} \text{ and from (a) the tangent}$$

slope is $-\frac{1}{x^2}$ So QP is $\perp$ to the

curve iff: $m_{QP} \cdot \left(-\frac{1}{x^2}\right) = -1 \Leftrightarrow \frac{1 - 2x}{x^2} = x$

$\Leftrightarrow x^4 + 2x - 1 = 0$ So it's at closest point $QP \perp$ curve